ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MATHEMATICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mathematics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods.

QUESTION 1

1.1.1 $x = 3y - 6$ ✓; $f^{-1}(x) = \frac{x+6}{3}$ ✓ (2)

1.1.2 $\frac{3+6}{3} = 3$ ✓ (1)

1.2.1 $x = 2y^2$ ✓; $y^2 = \frac{x}{2}$ ✓; $g^{-1}(x) = \sqrt{(\frac{x}{2})}, x \geq 0$ ✓ (3)

1.2.2 Without the restriction *g* is many-to-one ✓, so its inverse would not be a function; restricting the domain makes the inverse a function ✓. (2)

1.3 $y = 3^x$ through (0 ; 1), increasing ✓; $y = \log_3 x$ through (1 ; 0), increasing ✓; graphs symmetrical about $y = x$ ✓; line $y = x$ drawn ✓ (4)

1.4.1 $\log_3 27 = 3$ ✓ (1)

1.4.2 $0 < x < 1$ ✓✓ (2)

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QUESTION 2

2.1 $\lim \frac{(x-3)(x+3)}{x-3} = \lim (x+3)$ ✓ $= 6$ ✓ (2)

2.2 $f(x+h) = 2(x+h)^2 - 3 = 2x^2 + 4xh + 2h^2 - 3$ ✓; $f(x+h) - f(x) = 4xh + 2h^2$ ✓; $f'(x) = \lim_{h \to 0} \frac{4xh + 2h^2}{h}$ ✓ $= \lim_{h \to 0} (4x + 2h)$ ✓ $= 4x$ ✓ (5)

2.3.1 $y = 4x^3 - 2x^{-1} + x^{1/2}$ ✓; $\frac{dy}{dx} = 12x^2$ ✓ $+ 2x^{-2}$ ✓ $+ \frac{1}{2}x^{-1/2}$ ✓ (4)

2.3.2 $y = 4x^2 - 4x + 1$ ✓; $\frac{dy}{dx} = 8x - 4$ ✓ (2)

2.4 $f(3) = -2$ ✓; $f'(x) = 2x - 4$, $m = 2$ ✓; $y + 2 = 2(x - 3)$ ✓; $y = 2x - 8$ ✓ (4)

2.5.1 $g'(x) = 3x^2 - 6x - 9 = 0$ ✓; $3(x-3)(x+1) = 0$ ✓; $x = 3$ or $x = -1$ ✓; turning points $(-1 ; 10)$ ✓ and $(3 ; -22)$ ✓ (5)

2.5.2 $g''(x) = 6x - 6 = 0$, $x = 1$ ✓; point of inflection $(1 ; -6)$ ✓ (2)

2.5.3 Cubic shape (positive leading coefficient) ✓; turning points ✓; point of inflection ✓; *y*-intercept (0 ; 5) ✓ (4)

2.6.1 Length parallel to the river $= 120 - 2x$ ✓; $A = x(120 - 2x) = 120x - 2x^2$ ✓ (2)

2.6.2 $A' = 120 - 4x = 0$ ✓; $x = 30$ m, length $= 60$ m ✓; $A = 30 \times 60 = 1\,800$ m^2 ✓ (3)

2.7 $v = s'(t) = 10t + 2$ ✓; $v(3) = 32$ m/s ✓ (2)

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QUESTION 3

3.1 $i = \frac{0{,}1}{12}$, $n = 60$ ✓; $\text{F} = 800 \times \frac{(1 + 0{,}1 \div 12)^{60} - 1}{0{,}1 \div 12}$ ✓✓✓ = R61 949,66 ✓ (5)

3.2.1 $250\,000 = \frac{x[1 - (1{,}01)^{-60}]}{0{,}01}$ ✓✓; x = R5 561,11 ✓✓ (4)

3.2.2 $60 \times 5\,561{,}11 = \text{R}333\,666{,}60$ ✓; interest = R83 666,60 ✓ (accept R83 666,72 using the unrounded payment) (2)

3.2.3 Balance = present value of the remaining 36 payments ✓: (4)

$\frac{5\,561{,}11[1 - (1{,}01)^{-36}]}{0{,}01}$ ✓✓ = R167 431,21 ✓

3.3 Future-value annuity: equal regular payments are saved to build up an amount in the future ✓, e.g. a retirement or savings plan; present-value annuity: equal regular payments repay an amount received now ✓, e.g. a car loan or bond (2)

3.4 $(1 + \frac{0{,}11}{12})^{12} - 1$ ✓ = 11,57% ✓ > 11,4%, so 11% compounded monthly is better ✓ (3)

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QUESTION 4

4.1 $\cos(45° + 30°) = \cos 45° \cos 30° - \sin 45° \sin 30°$ ✓ $= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2}$ ✓✓ $= \frac{\sqrt{6} - \sqrt{2}}{4}$ ✓ (4)

4.2.1 $\tan 28° = \frac{45}{\text{FP}}$ ✓; $\text{FP} = \frac{45}{\tan 28°}$ ✓ = 84,63 m ✓ (3)

4.2.2 $\text{PQ}^2 = 84{,}63^2 + 60^2 - 2(84{,}63)(60)\cos 110°$ ✓✓; PQ = 119,32 m ✓✓ (4)

4.3 $1 - 2\sin^2 x = \sin x$ ✓; $2\sin^2 x + \sin x - 1 = 0$; $(2\sin x - 1)(\sin x + 1) = 0$ ✓; $\sin x = \frac{1}{2}$: $x = 30°$ or $150°$ ✓; $\sin x = -1$: $x = 270°$ ✓ (4)

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QUESTION 5

5.1.1 $(x - 2)^2 + (y + 3)^2 = 25$ ✓✓ (2)

5.1.2 $(5 - 2)^2 + (1 + 3)^2 = 9 + 16$ ✓ = 25 ✓, so P lies on the circle (2)

5.1.3 $m_{\text{MP}} = \frac{1 + 3}{5 - 2} = \frac{4}{3}$ ✓; $m_{\text{tangent}} = -\frac{3}{4}$ ✓; $y - 1 = -\frac{3}{4}(x - 5)$ ✓; $y = -\frac{3}{4}x + \frac{19}{4}$ ✓ (4)

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5.2.1 Mean $= 40 \div 5 = 8$ ✓; deviations$^2 = 16; 0; 4; 4; 16$ ✓; sum $= 40$ ✓; $\sigma = \sqrt{(}$ $\frac{40}{5}$) ✓ $= 2{,}83$ ✓ (accept s $= 3{,}16$ using n $- 1$) (5)

5.2.2 Interval 5,17 to 10,83 ✓: 6, 8 and 10, i.e. 3 values ✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE